



54-port 10G Core Routing Switch (NAV-C48S2Q-4Q)



High-Speed Layer 3 Industrial Connectivity

The NAV-C48S2Q-4Q is a high-performance L3 managed Ethernet switch designed for metropolitan area networks, data centers, and campus networks. It features 48x 1/10G SFP+ fiber ports, 2x 40G QSFP28 fiber ports, and 4x 100G QSFP28 fiber ports to effectively expand network capacity.



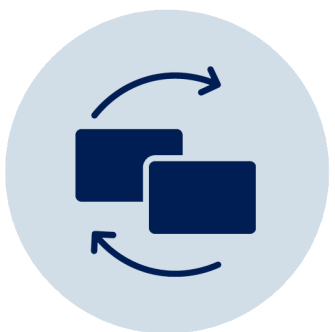
Advanced Hardware

Built on advanced hardware architecture with a high-performance ASIC chip and multi-core processor, it offers a switching capacity of up to 1.96Tbps, catering to high-density data center needs.



Stable & Reliable

Features the HPS (Hitless Protection System) with redundant, hot-swappable power supplies. Supports ISSU (In-Service Software Upgrade) & ring network protection to ensure uninterrupted data forwarding.



Rich Features

Comprehensive L3 routing protocols, including RIP, OSPF, BGP, and MPLS VPN. Full support for IPv6 tunneling and multicast routing makes it ideal for HD video and large-scale enterprise networks.

Switching Capacity

1.96Tbps

Forwarding Rate

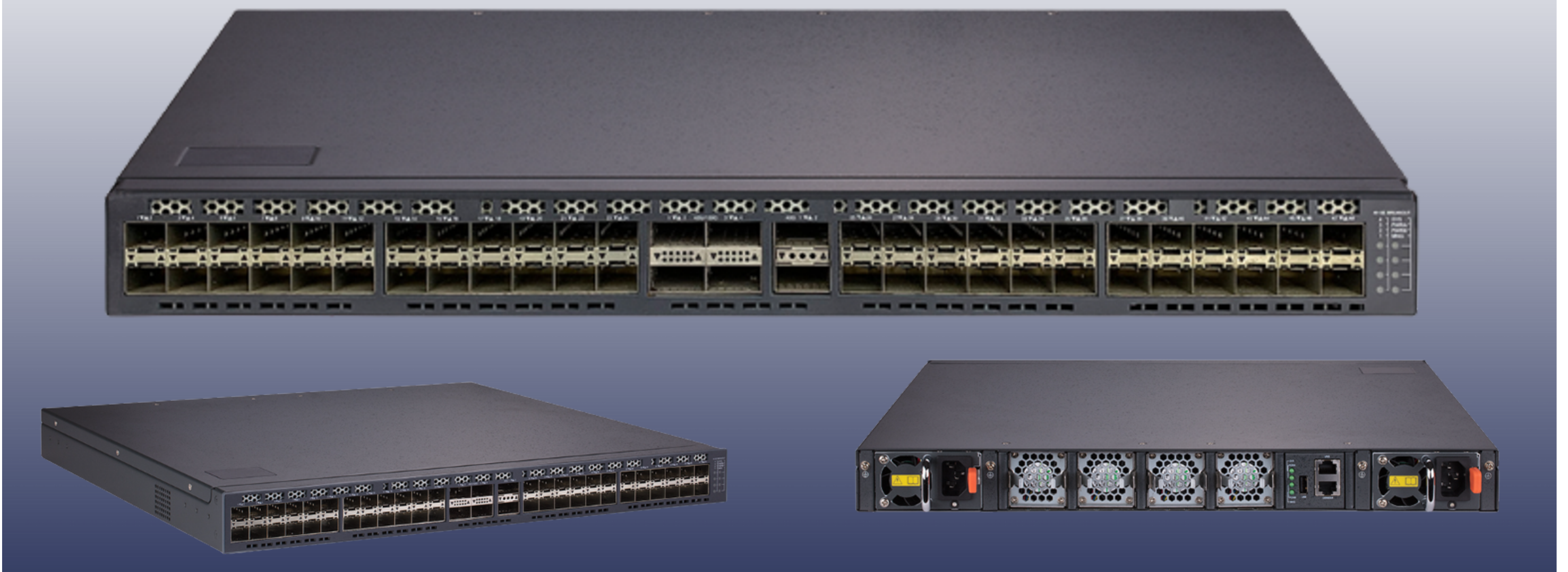
1428Mpps

Core Routing Protocol

L3 Managed

This switch integrates network security, traffic analysis, and virtualization services. It provides flexible airflow management and complies with IEEE 802.3az Energy Efficient Ethernet standards to reduce operational costs while maintaining peak performance.

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Product Overview

The **NAV-C48S2Q-4Q** is a high-performance Layer 3 (L3) managed Ethernet switch designed to meet the rigorous demands of metropolitan area networks (MANs), data centers, campus networks, and large enterprise environments.

Designed for high-density applications, this 1U/19-inch rack-mountable unit features 48x 1/10G SFP+ fiber ports, 2x 40G QSFP28 fiber ports, and 4x 100G QSFP28 fiber ports.

The NAV-C48S2Q-4Q offers comprehensive L3 management functions, integrating IPv6, MPLS VPN, network security, traffic analysis, and virtualization. It provides high-performance L2/L3/L4 line-speed switching and employs advanced data center reliability technologies—such as uninterrupted upgrades, continuous forwarding, and redundant protection—to ensure maximum network uptime. With support for advanced routing protocols like RIP, OSPF, BGP, and PIM-DM/SM, it is the ideal choice for both traditional and fully virtualized data centers.

Key Features

High-Density Core Architecture

- Features 48× 1/10G SFP+, 2× 40G QSFP28, and 4× 100G QSFP28 fiber ports.
- Delivers 1.96Tbps switching capacity via high-performance ASIC chips for non-blocking data center cores

Continuous Reliability

- Supports ISSU (In-Service Software Upgrade) for uninterrupted data forwarding during updates.
- Hitless Protection System (HPS) with hot-swappable, redundant power ensures seamless failover.

Advanced L3 Services

- Full IPv6 support, including tunneling (manual, 6to4, ISATAP) and neighbor discovery.
- MPLS VPN and comprehensive multicast routing (PIM-DM/SM, OSPF, BGP) for large-scale networks.

Robust Security Control

- Hardware-level packet filtering defends against DDoS, TCP SYN, and UDP flood attacks.
- Supports IEEE 802.1x, Radius, Tacacs+, and uRPF reverse routing lookup for strict access control.

Intelligent O&M

- Real-time CPU/memory monitoring, Ping detection, and cable diagnostics.
- Secure management via HTTPS, SSH V2, and SNMP v1/v2/v3 with RMON traffic statistics.

A high-performance L3 core routing switch with 48x 10G SFP+ ports and 100G uplinks. Built for data centers, it offers 1.96Tbps capacity, virtualization support, and carrier-grade redundancy for zero downtime.

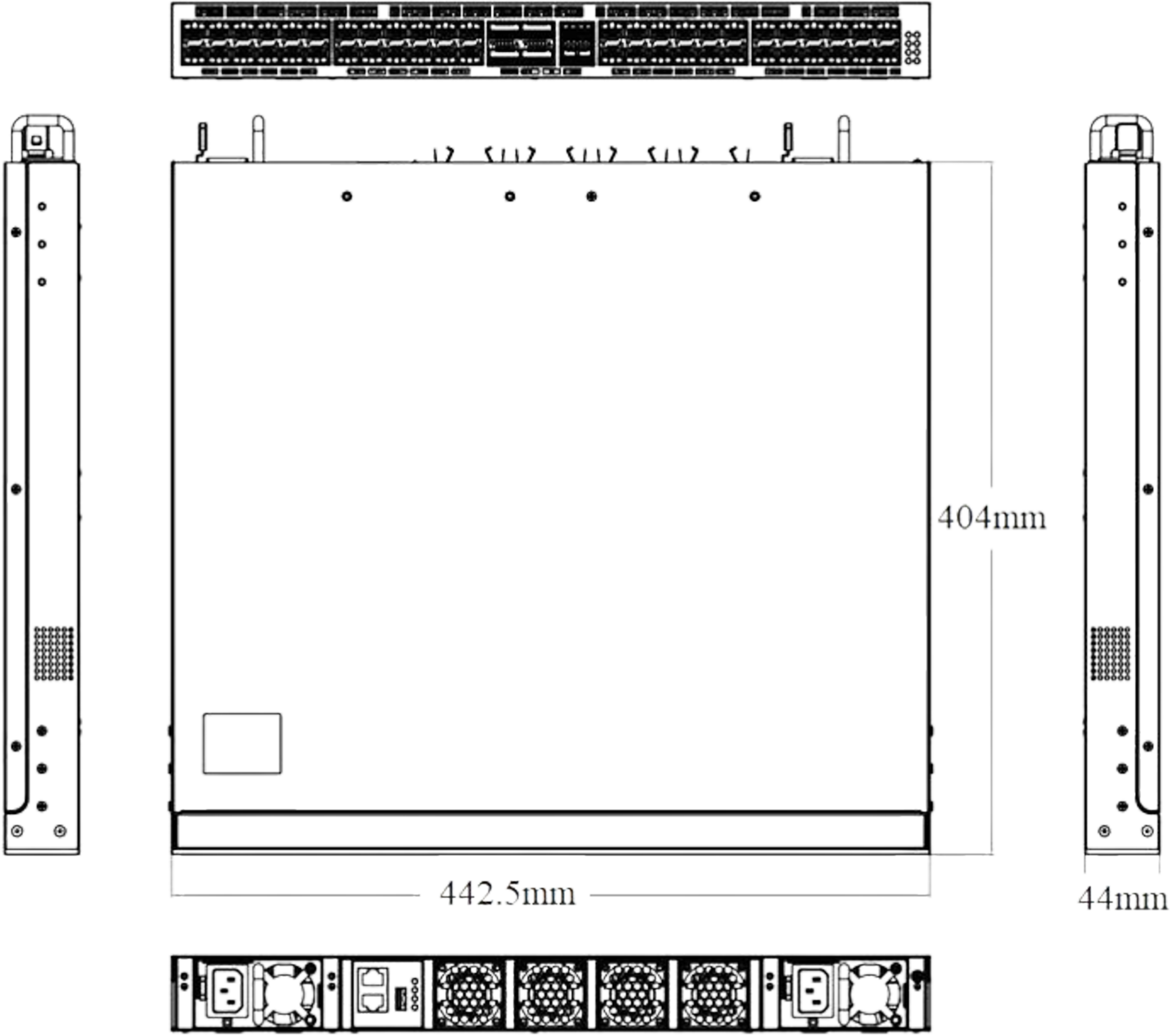
Technical Specification

Feature	Specification
Model	NAV-C48S2Q-4Q 1
Interface Characteristics	
Fixed Port	1RS232 Console port (9600,8,N,1) 2 48*1/10G SFP+ fiber ports (Data) 3 2*40G QSFP28 uplink fiber ports (Data) 4 4*100G QSFP28 uplink fiber ports (Data) 5 1USB 2.0 configuration port 6
Optical Fiber Port	10G SFP+/QSFP28 optical fiber ports, default no include optical modules (optional single-mode/ multi-mode, single fiber/ dual fiber optical module. LC) 7 10G SFP+ optical fiber port can be backward compatible with 1G optical fiber module. 8
Optical Cable/ Distance	Multi-mode: 850nm/0-550m, Single-mode: 1310nm/0-40km, 1550nm/ 0-120km. 9
Chip Parameter	
Network Management Type	L3 10
Network Protocol	IEEE 802.3u 100Base-TX, IEEE 802.3ab 1000Base-T, IEEE 802.3z 1000Base-X, IEEE 802.3ae 10Gb/s Ethernet, IEEE 802.3x 11
Forwarding Mode	Store and Forward (Full Wire Speed) 12
Switching Capacity	1.96Tbps (non-blocking) 13
Forwarding Rate@64byte	1428Mpps 14
MAC	64K 15
Buffer Memory	64M 16
Jumbo Frame	16K 17
LED Indicator	Power: PWRA, PWRB (Green), System: SYS (Green), Network management: MNG (Green) 18
Power Supply	
Total PWR/ Input Voltage	2*480W/ (AC100-240V) 19
Power Consumption	Standby<50W, Full Load<160W 20
Power Supply	Built-in power supply, AC100~240V 50-60Hz, 2*5.0A 21
Physical Parameter	
Operation Temp/ Humidity	-20°C~+55°C, 5%~90% RH Non condensing 22
Storage Temp/Humidity	-40°C~+75°C, 5%~95% RH Non condensing 23
Dimension (L*W*H)	442.5*404*44mm 24
Net/Gross Weight	<6.8kg /<7.0kg 25
Installation	Desktop, 1U/19" cabinet 26
Certification & Warranty	
Lightning Protection	Port lightning protection: 6KV 8/20us, Protection level: IP30 27
Certification	CCC, CE mark, commercial, CE/LVD EN62368-1, FCC Part 15 Class B, RoHS 28
Warranty	3 years, lifelong maintenance. 29
Network Management Feature	
Stacking	Virtualization, Local stacking and remote stacking 30 Stacking via standard Ethernet interface 31 MAD stack split detection mechanism based on LACP, BFD, and ARP 32 Distributed device management, distributed link aggregation, and distributed elastic routing 33
IPv4	Policy routing, BFD for OSPF, BGP 34 Equal-cost routing to achieve load balancing 35 Static routing, RIP v1/v2, OSPF, BGP, IS-IS, BEIGRP 36
IPv6	MLD v1/v2, MLD Snooping 37 ICMPv6, DHCPv6, ACLv6, IPv6 Telnet 38 Manual tunnel, ISATAP tunnel, 6to4 tunnel 39 IPv6 static routing, RIPng, OSPFv3, BGP4+ 40 IPv6 neighbor discovery, Path MTU discovery 41

MAC Switching	View and clear the MAC address 42 MAC address aging time is configurable 43 IEEE 802.1AE MacSec Security Control 44 Limit the number of MAC address learning 45 Black hole MAC table entry, MAC address filtering function 46 Static configuration and dynamic learning of MAC addresses 47
VLAN	Basic QinQ and flexible QinQ function 48 4K VLAN entries, GVRP, Private VLAN, 1:1 and N:1 VLAN Mapping 49
Link Aggregation	10GE port aggregation, Static aggregation, Dynamic aggregation 50
Flow Monitoring	SFLOW 51
DHCP	DHCP Client, DHCP Snooping, DHCP Relay, DHCP Server 52 DHCP Snooping option82/ DHCP Relay option82 53 Zero configuration methods such as DHCP auto-config and CWMP-TR069 54
STP/ERPS	802.1D (STP), 802.1W (RSTP), 802.1S (MSTP) 55 BPDU protection, Root protection, Loop protection 56 ERPS Ethernet Ring Protection Protocol (G.8032) 57
Multicast	Multicast group policy and multicast number limit 58 IGMP V1/v2/v3, IGMP Snooping, IGMP Fast Leave 59 PIMSM.PIM--DM, Multicast traffic cross VLAN duplication 60
ARP	Static entries, ARP source suppression 61 Standard proxy ARP and local proxy ARP 62 Free ARP, Dynamic ARP Inspection, ARP anti-attack 63 ARP Detection (check based on DHCP Snooping security entries, 802.1x entries, or IP/MAC static binding entries) 64
Mirroring	Flow mirroring, N:4 port mirroring, Local and remote port mirroring 65
MPLS VPN	MCE, MPLS TE, MPLS OAM, LDP protocol, P/PE of MPLS VPN 66
QoS/ACL	Traffic supervision and traffic shaping 67 SP, WRR, SP+WRR scheduling mode 68 CAR traffic restriction, 802.1P/DSCP priority re-marking 69 Tail-Drop, WRED, and other congestion avoidance mechanisms 70 Traffic classification based on each field of the L2/L3/L4 protocol header 71 Ingress and Egress ACL, matching L2, L3, L4 and IP five-tuple, copying, forwarding, and discarding 72 Hash same-source and same-destination load balancing to ensure session integrity of traffic output 73 Identification and filtering of L2/L3/L4 based ACL 74
Security	Urpf, Port isolation, Port security, IP + MAC + port binding 75 IEEE 8021x certification, DHCP Snooping, DHCP Option 82 76 Radius and BDTacacs+, Command line hierarchical protection 77 Suppression of broadcast, multicast, and unknown unicast packet 78 Defend against DDoS attack, SYN Flood attack of TCP, and UDP Flood attack 79

Reliability	EAPS, ERPS ring network protection 80 ISSU service without interruption system upgrade 81 HSRP, VRRP hot standby protocol, GR for OSPF, BGP 82 Optional power supply 1+1 backup, BFD for OSPF, BGP 83 Static/LACP link aggregation, support cross-service card link aggregation 84
Management	File upload and download management in TFTP mode 85 Telnet remote maintenance, ZTP(Zero Touch Provisioning) 86 Power alarm, Fan, temperature alarm, Console, Telnet, SSH 2.0 87 ISSU, Track, Tracert, SFLOW and other traffic statistics analysis 88 NTP, Ping, Debug information output, Web browser management 89 System logs, Graded alarm, SNMP v1/v2/v3, 802.1AG and 802.3AH 90 RMON event history, SNMP (Simple Network Management Protocol) 91
Energy Saving	IEEE802.3az green energy Ethernet 92
System	Web browser: Mozilla Firefox 2.5 or higher, Google Chrome V42 or higher, Cat5 and above Ethernet cable 93 TCP/IP, network adapter, and network operating system (such as Microsoft Windows, Linux, Mac OS X) installed on each computer in the network 94

Dimensions



Ordering Information

Model	Description	Power Supply
NAV-C48S2Q-4Q	L3 managed Ethernet core routing switch with 48x 1/10G SFP+ fiber ports, 2x 40G QSFP28 fiber ports, and 4x 100G QSFP28 fiber ports. Supports redundant dual AC power supply and standard 1U/19" cabinet installation. (Note: Optical modules are not included and must be purchased separately.) 55	2x 480W
Note: The optical module is not included and needs to be purchased.		

Packing List

Content	Quantity	Unit
NAV-C48S2Q-4Q (54-port 10G core routing switch)	1	Set
AC Power Cable	1	PC
Mounting Kit	1	Set
Warranty Card and Certificate of Conformity	1	PC

Optical Module

Type	Model	Description
1.25G SFP	2630	Multi-mode, Dual Fiber, 850nm, 550m, LC
	2632	Single-mode, Dual Fiber, 1310nm, 20km, LC
	2612-T	Single-mode, Single Fiber, TX1310nm/RX1550nm, 20km, LC
	2613-R	Single-mode, Single Fiber, TX1550nm/RX1310nm, 20km, LC
10G SFP+	6630	Multi-mode, Dual Fiber, 850nm, 300m, LC
	7832	Single-mode, Dual Fiber, 1310nm, 20km, LC
	7832-33	Single-mode, Single Fiber, TX1330nm/RX1270nm, 20km, LC
	7832-27	Single-mode, Single Fiber, TX1270nm/RX1330nm, 20km, LC